

Diagnosing Where Global Boundary Models Fail on SEA Smallholdings

BUCKET 5 - FIELD / PARCEL BOUNDARY AND SUB-FIELD STRUCTURE

For agricultural extension agronomists supporting smallholder farmers in Vietnam and Cambodia, we run a stratified diagnosis of exactly which landscape conditions (field size, slope, tree-canopy overlap, fragmentation) break the zero-shot global field-boundary model Delineate Anything v2, then fix the single dominant failure mode with a terrain/canopy-conditioned adapter and measure the gain specifically on the strata where the global model fails.

WEAK

The imagery/geography/budget/pitch gates all pass cleanly, but the H1 novelty case is genuinely fragile: the specific failure mode this candidate needs to discover and fix is being actively and rapidly picked apart by several adjacent 2024-2026 papers from a small number of highly active research groups, and the flagship SEA dataset (AI4SmallFarms) may not even contain the terrain/canopy variance the pitch's framing assumes. This candidate should proceed only if the week-1 kill-criteria confirm a real, causally demonstrable, stratum-specific failure with sufficient variance in the available data; absent that confirmation, it risks collapsing into exactly the generic 'applied X to Y region' pattern the brief flags it for.

Confidence: Medium. Would rise to High after (a) actually running the frozen Delineate Anything v2 model on AI4SmallFarms and computing the stratified breakdown — this is precisely the week-1 kill-criteria check and would immediately confirm or refute the premise — and (b) reading GTPBD-MM's full method and results sections (not just the abstract) to confirm whether its geographic coverage or ablations already extend to SEA-like canopy-overlap conditions.

Hard constraint check

Constraint	Verdict	Reasoning
H1 • Novelty	CAVEAT	WEAK/CONDITIONAL PASS — genuine novelty exists only if the week-1 diagnosis confirms a real, unaddressed, and causally demonstrable canopy/terrain-driven failure mode specific to SEA; given the density of adjacent 2024-2026 fixes for fragmentation, cloud cover, and generic terrain, this is the more fragile of the two candidates on H1.
H2 • Not generic	SEE NOTE	MARGINAL PASS — not a literally banned pattern, but structurally close to the 'apply global model X, find it fails in region Y, fine-tune' genericity trap the brief explicitly names; only cleared if the mechanism diagnosis is rigorously causal.
H3 • Imagery is primary input	PASS	Sentinel-2 is the primary imagery input; DEM and WorldCover auxiliary channels, resolutions, revisit, and licences all stated.
H4 • Geography & user	PASS	Vietnamese/Cambodian smallholder-support agronomists and co-ops are named; SEA is an explicitly acceptable geography per H4 even without a Singapore-specific angle.
H5 • Compute & budget	PASS	dataset size (<20GB), compute (a few hours, fits in 24GB), and cash cost (~RM0) are all comfortably within budget.

The problem

Global field-delineation foundation models now report strong aggregate zero-shot accuracy across 61-100 countries (Delineate Anything v2, arXiv:2607.19069), but country- or continent-level aggregates hide stratum-specific collapse: sub-0.5ha plots, sloped/terraced terrain, and tree-crop/agroforestry canopy overlap are known to break flat-farmland-trained boundary models (as GTPBD/GTPBD-MM, arXiv:2507.14697 and arXiv:2604.12315, showed for terraced parcels generally). Southeast Asian smallholdings combine several of these hard conditions simultaneously, yet no published evaluation decomposes performance by these strata for SEA specifically, and no fix has been validated against the specific mechanism that breaks the model there (as opposed to a general fine-tune-on-local-data gain). Extension agronomists therefore have no evidence-based way to know whether a deployed global model's outputs can be trusted on a given smallholder plot, or why it fails when it does.

Proposed technical approach

(1) Build a stratified SEA smallholder evaluation set from AI4SmallFarms (Vietnam and Cambodia, 439,001 manually digitised field polygons, IEEE GRSL 2023), tagging every tile with: field-size quartile (from the polygons themselves), slope/aspect bucket (from Copernicus DEM GLO-30), tree-canopy-cover fraction bucket (from ESA WorldCover 10m tree-cover class as a canopy-overlap proxy), and a fragmentation index (parcel density per km²). (2) Run Delineate Anything v2 zero-shot (frozen, pretrained weights) on every tile and compute per-tile instance-segmentation metrics (mAP@0.5, boundary IoU, object-F1); decompose variance in per-tile performance across the four stratum axes (e.g. gradient-boosted feature importance / ANOVA) to identify which axis is the dominant, statistically significant driver of failure. (3) Implement a targeted architectural fix for the dominant axis only: most plausibly a terrain-aware adapter (slope/aspect as an auxiliary input channel fused into the frozen backbone via a lightweight cross-attention adapter, following the DEM-conditioning recipe validated by GTPBD-MM's ETerra for terraced parcels) or, if canopy closure dominates, a WorldCover tree-cover channel plus an auxiliary canopy-decontamination loss that discourages merging adjacent under-canopy parcels into one instance. (4) Fine-tune only the adapter and decoder head (backbone frozen) on a small labelled SEA subset; evaluate the fix specifically on the failing stratum against two controls: the frozen zero-shot baseline, and a naive full fine-tune on the same SEA subset with no terrain/canopy conditioning at all (this ablation isolates whether any gain is due to the specific conditioning mechanism or is simply generic domain adaptation from seeing local data).

Novelty

WHAT THE DELTA ACTUALLY IS

The defensible delta, if it holds, is the causal mechanism diagnosis (which specific stratum — size, slope, canopy, fragmentation — actually drives failure, isolated by ablation against naive fine-tuning) combined with a measured, stratum-targeted gain over the frozen global SOTA. This is explicitly not 'applied X to Y country': the deliverable is a falsifiable, quantified failure mechanism plus an architectural fix validated against a domain-adaptation confound, not a re-run of an existing pipeline on new geography. However, this delta must be stated honestly as narrow and time-sensitive: as of this research pass, at least three adjacent 2024-2026 papers have already closed parts of the failure space this candidate targets — 'Fields of the Planet' (arXiv:2607.04449) already fixes the small-field/fragmentation failure mode via higher-resolution (3m PlanetScope) imagery rather than terrain/canopy conditioning; PTAiT3D-CA (arXiv:2409.13568) already validates cross-region transfer including on AI4SmallFarms itself using SAR+optical fusion to address cloud contamination, a major real SEA confound; and GTPBD-MM/ETerra (arXiv:

2604.12315) already validates DEM+text conditioning for terraced parcels, just not for SEA or for canopy overlap specifically. The remaining defensible novel territory is narrow: canopy-closure/agroforestry-specific conditioning combined with SEA smallholder terrain, which as far as this research could confirm has not yet been done — but this gap could close within months given the publication cadence of the FTW and Delineate Anything research groups.

PRIOR ART THAT THREATENS IT

Delineate Anything v2 (arXiv:2607.19069) itself already reports a 100-country zero-shot benchmark, though at country-aggregate, not stratum, granularity. 'The first global agricultural field boundary map at 10m resolution' (arXiv:2605.11055, May 2026) already flags over-fragmentation in smallholder systems as a known, named, unresolved failure mode — meaning the fragmentation half of this candidate's diagnosis is not new. 'Fields of the Planet' (arXiv:2607.04449, July 2026) already demonstrates that the small-field/fragmentation failure is substantially fixed by higher-resolution imagery (PQ on sub-0.5ha fields rising from 5.8 to 15.7 at 3m vs 10m), which directly undercuts a 'fragmentation is the dominant, terrain/canopy-fixable failure mode' framing if fragmentation turns out to dominate in week 1's diagnosis. PTAViT3D-CA (arXiv:2409.13568, tested explicitly on AI4SmallFarms) already shows SAR+optical fusion resolves cloud-contamination-driven failures across smallholder regions including this candidate's own target dataset. GTPBD-MM/ETTerra (arXiv:2604.12315, April 2026) already validates the DEM-conditioning recipe this candidate proposes to reuse, for terraced (though non-SEA) parcels. PRUE (arXiv:2603.27101, March 2026) already runs a systematic 18-model evaluation of segmentation and geospatial foundation models on the FTW benchmark for robustness under real-world conditions, partially closing the 'systematic stratified diagnosis' novelty space, though not with SEA/canopy-specific strata. Taken together, this is an unusually dense cluster of directly adjacent 2024-2026 work from a small number of active groups (the Microsoft/ASU Fields-of-The-World team; the Lavreniuk/Kussul Delineate Anything team; the GTPBD/GTPBD-MM team), each shipping roughly every 1-2 months in 2026.

GENERIC-PATTERN CHECK

On its face the core is not one of the eight banned H2 patterns (not a chatbot, dashboard, leaf-disease classifier, NDVI viewer, tabular yield model, crop-type-classification rerun, or API-wrapper app). However, the brief itself flags this candidate as the highest genericity risk in the batch, and that flag is justified: the one-sentence template — 'a global model underperforms in region Y because of local condition X, so we fine-tuned it with X as an extra input' — is a pattern that could equally describe dozens of other remote-sensing FYPs (flood mapping, building-footprint extraction, land-cover classification, all adapted from a global model to a local condition). This candidate only clears H2 if the diagnosed mechanism is genuinely causal and falsifiable (e.g. a specific, testable claim such as 'overlapping tree crowns cause the instance-segmentation head to merge adjacent parcel masks, verifiable by visual inspection of merged-instance error cases and by an ablation that removes canopy overlap synthetically') rather than a generic 'accuracy went up after we added more local data and an extra channel' result. Verdict: marginal pass, conditional on the mechanism claim being rigorously causal, not merely correlational.

Sensors and imagery

Sensor	Resolution	Revisit (days)	Bands	Licence
Sentinel-2 MSI	10 m	5	RGB+NIR composite (matching Delineate Anything v2 / FTW training convention)	Copernicus free and open data licence
Copernicus DEM GLO-30 (auxiliary, static terrain conditioning channel)	30 m	static (one-time acquisition, 2010-2015 radar mission)	elevation (derive slope/ aspect)	Free, open licence (Copernicus/ESA/DLR); no login for STAC access

ESA WorldCover (auxiliary, static canopy-proxy conditioning channel)	10 m	annual product (2020/2021 epochs), used as static proxy	tree-cover class fraction	CC BY 4.0
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Sentinel-2 optical imagery is the primary segmentation input; DEM and WorldCover are auxiliary static conditioning channels fused into the model, not substitutes for imagery.

Yes. Sentinel-2 imagery (matching the training convention of the global model being diagnosed) is the primary input to the segmentation model. DEM and WorldCover are auxiliary conditioning channels used to explain and fix failures, analogous to GTPBD-MM's own DEM-conditioning design, and do not replace imagery as the core input.

Datasets

AI4SMALLFARMS

Access route	https://doi.org/10.17026/dans-xy6-ngg6 (DANS data repository); paper in IEEE Geoscience and Remote Sensing Letters
Auth required	No
Licence	Open dataset via DANS; exact licence string not independently re-verified in this pass [uncertain]
Approx. size	439,001 field polygons across 62 tiles of ~5x5km at Sentinel-2 10m resolution — vector + raster together estimated at a few GB
Accessible from MY	Yes (public DANS repository, no geofencing found)
How verified	WebSearch confirming IEEE GRSL 2023 publication (IEEE Xplore document 10278130) and the DANS DOI download route

DELINEATE ANYTHING V2 PRETRAINED WEIGHTS + FBIS-73M REFERENCE

Access route	https://github.com/Lavreniuk/Delineate-Anything (code, pretrained weights, dataset links)
Auth required	No
Licence	Public GitHub release; specific licence terms not independently re-verified [uncertain]
Approx. size	Pretrained weights: hundreds of MB (YOLOv11-based backbone per the companion Delineate Anything Flow paper); FBIS-73M full dataset not required, only the released weights are needed
Accessible from MY	Yes
How verified	arXiv abstract (2607.19069) confirms public release of code, weights and dataset at the stated GitHub URL

COPERNICUS DEM GLO-30

Access route	Microsoft Planetary Computer STAC collection 'cop-dem-glo-30'; also AWS Registry of Open Data and OpenTopography
Auth required	No
Licence	Free, open (DLR/Airbus/ESA)
Approx. size	Negligible for the AI4SmallFarms footprint (a few tiles, <1GB)
Accessible from MY	Yes
How verified	WebSearch confirming Planetary Computer STAC endpoint and AWS Registry of Open Data listing

ESA WORLDCOVER

Access route	Microsoft Planetary Computer STAC collection 'esa-worldcover'
Auth required	No
Licence	CC BY 4.0
Approx. size	Negligible for the AI4SmallFarms footprint (<1GB)
Accessible from MY	Yes [uncertain: not independently re-verified via a fresh WebFetch in this pass, based on well-established prior knowledge of this standard open dataset]
How verified	Not independently re-fetched in this research pass; flagged uncertain and should be confirmed in week 1 alongside the DEM check

TOTAL DATA VOLUME

Well under 20GB total (AI4SmallFarms imagery+labels, pretrained model weights, and DEM/WorldCover tiles clipped to the same small footprint). Comfortably within the H5 practical ceiling.

LABELS AND GROUND TRUTH

AI4SmallFarms provides real, manually-digitised field-boundary polygons for Vietnam and Cambodia — genuine ground truth for both the stratified evaluation and the fine-tuning step, not a proxy. Slope and canopy-cover strata are derived from free auxiliary raster layers (DEM, WorldCover), requiring no new manual labels. Risk, stated plainly: AI4SmallFarms' actual footprint (Mekong-delta-adjacent Vietnam and Cambodia) may not contain much genuine terracing or dense agroforestry canopy overlap — if so, the terrain/canopy strata this candidate needs to diagnose may have insufficient variance in the available labelled data, and a supplementary small hand-digitised set (e.g. over Philippine or Indonesian terraces) would be needed, which is itself a real, unbudgeted labelling task.

Baselines to beat

Paper	Reported metric	Dataset
Delineate Anything v2: A Global Foundation Model for Field Delineation arXiv preprint 2026 · arXiv: 2607.19069	+0.284 mAP@0.5 (+103.3% relative) over predecessor Delineate Anything, on a 100-country zero-shot benchmark (aggregate, not stratified)	FBIS-73M (73M instances, 61 countries)

Delineate Anything: Resolution-Agnostic Field Boundary Delineation on Satellite Imagery arXiv preprint 2025 · arXiv: 2504.02534	+88.5% mAP@0.5, +103% mAP@0.5:0.95 over prior state of the art	FBIS-22M
Fields of The World: A Machine Learning Benchmark Dataset for Global Agricultural Field Boundary Segmentation AAAI 2025 · arXiv:2409.16252	Zero-shot and fine-tuning performance across 24 countries; better held-out-country zero-shot performance for models pretrained on diverse data	FTW (70,462 samples, 24 countries)
GTPBD-MM: A Global Terraced Parcel and Boundary Dataset with Multi-Modality (ETerra baseline) arXiv preprint 2026 · arXiv: 2604.12315	Demonstrates DEM+text conditioning improves terraced-parcel delineation coherence over image-only baselines (China-centric terraces, not SEA)	GTPBD-MM (built on GTPBD, 200,000+ terraced parcels)
AI4SmallFarms: A Dataset for Crop Field Delineation in Southeast Asian Smallholder Farms IEEE Geoscience and Remote Sensing Letters 2023 · IEEE Xplore 10278130	U-Net/FracTAL-style baseline field delineation performance on Vietnam/Cambodia smallholder tiles	AI4SmallFarms (439,001 polygons, 62 tiles)

Evaluation protocol

Per-stratum instance-segmentation metrics (mAP@0.5, boundary IoU, object-F1) computed on held-out AI4SmallFarms tiles, broken out by field-size quartile, DEM-slope bucket, WorldCover tree-cover-fraction bucket, and parcel-density (fragmentation) bucket. Statistical test: ANOVA or gradient-boosted feature-importance across the four stratum axes on per-tile zero-shot performance, to identify which axis explains the most variance (the 'dominant failure mode'). Success criterion: a paired, per-tile, statistically significant IoU/F1 gain on the worst-performing stratum after the targeted terrain/canopy fix, over both (a) the frozen Delineate Anything v2 baseline and (b) a naive-fine-tune-with-no-conditioning ablation on the same SEA subset — the second comparison is essential to prove the gain comes from the specific mechanism diagnosed, not merely from local fine-tuning.

Feasibility

COMPUTE

Peak memory	Modest, likely under 8GB: fine-tuning is restricted to a lightweight adapter plus decoder head on a frozen backbone; Delineate Anything's DelAny model is YOLOv11-based (lightweight, per the Delineate Anything Flow paper's own '400x faster than SAM2' inference claim)
Est. training hours (MPS)	Estimated 1-4 hours for adapter/decoder fine-tuning on the small AI4SmallFarms-derived labelled subset

Fits 24 GB	Yes, comfortably
MPS-compatible ops	YOLO-family backbones and standard conv/attention adapter layers are well supported on PyTorch MPS; lower risk than the custom SITS-encoder ops in the sub-field-zoning candidate
Colab/Kaggle fallback	No, but useful for a larger stratified-evaluation sweep if desired
Notes	The larger effort is the stratification pipeline (joining DEM/WorldCover rasters to AI4SmallFarms tiles and computing per-tile metrics), which is CPU-bound raster/vector work, not compute-intensive.
Cash cost	Approximately RM 0. AI4SmallFarms is a free DANS download, Delineate Anything v2 weights are freely released on GitHub, and DEM/WorldCover are free via Planetary Computer. An optional extension using commercial-grade PlanetScope/NICFI tropical-forest imagery for supplementary canopy-overlap examples was considered but is not required and its free-tier access route was not independently verified in this pass — flagged as skippable, budget RM0 if omitted.
Build : research	Approximately 60:40 engineering:research. Engineering: the DEM/WorldCover-to-tile stratification pipeline, the per-stratum metrics computation, and the adapter fine-tuning pipeline plus demo app. Research: the mechanism diagnosis itself, designing the ablation that separates 'the specific conditioning helped' from 'any local fine-tuning helps', and honestly positioning the work against the unusually dense adjacent 2024-2026 literature.
Publication venue fit	CVPR EarthVision or IGARSS workshop paper — failure-mode/robustness-diagnosis papers with a targeted fix fit these venues well. Remote Sensing (MDPI) is a realistic fallback given the applied SEA agriculture framing.

DEMO ARTIFACT

A laptop demo showing one SEA tile: frozen Delineate Anything v2 boundary predictions overlaid on the Sentinel-2 image (visibly merged/broken instances in the high-canopy-overlap or high-slope zone of the tile), side by side with the fixed model's predictions on the same tile, plus a bar chart of per-stratum IoU before and after the fix. Runs in under 2 minutes on pre-cached tiles and pre-trained weights.

User and pain

USER SEGMENT

Agricultural extension agronomists and smallholder-support cooperatives operating in Vietnam and Cambodia (matching AI4SmallFarms' actual coverage), who use field-boundary maps for parcel-level crop registration, targeted input-subsidy allocation, or (for tree-crop plots) deforestation/EUDR-style traceability checks. Extension to other SEA smallholder contexts (e.g. Malaysian or Indonesian oil-palm/cocoa smallholdings with heavier canopy overlap) is plausible but not currently backed by a verified labelled dataset in this research pass — stated as a limitation, not a claim.

QUANTIFIED PAIN

Delineate Anything Flow (arXiv:2511.13417) itself quantifies the scale of boundary-completeness failure in a fragmented/smallholder-adjacent system: for Ukraine, its improved pipeline delineated 3.75M fields (at 5m) versus 2.66M from an operational Sinergise Solutions product and 1.69M from NASA Harvest — a roughly 40-120% undercount by less-tuned pipelines when field fragmentation is high, illustrating the practical scale of the problem this candidate targets, though Ukraine is not itself an SEA smallholder analogue and this number is used only as an illustrative proxy for what boundary-completeness failure costs at scale. **[uncertain]** A specific, sourced number for the hectares or percentage of Vietnamese/

Cambodian smallholder cropland affected by canopy overlap or terracing specifically, or the direct cost to extension agronomists of unreliable global-model boundary outputs, was not found in the sources reviewed and would need primary quantification (e.g. from FAO or national agricultural statistics) in week 1.

Key risks

1. Data-sufficiency risk: AI4SmallFarms' actual footprint (Mekong-delta-adjacent Vietnam/Cambodia) may not contain enough genuine terracing or dense canopy-overlap variance to support the stratified diagnosis the pitch assumes — if strata don't vary meaningfully, the core premise collapses and would require supplementary hand-digitised data (an unbudgeted labelling task).
2. Novelty-erosion risk: the Fields-of-The-World and Delineate Anything research clusters are each shipping roughly one new closely-adjacent paper per 1-2 months through 2026 (Fields of the Planet, PRUE, the global 10m map, GTPBD-MM); a real probability that the exact SEA canopy/terrain diagnosis appears from one of these groups before this project concludes.
3. Mechanism/causality risk: 'accuracy dropped because of X' claims are easy to assert and hard to prove causally when resolution, terrain, canopy, and cloud cover are correlated in real SEA imagery — a weak, correlational mechanism claim would tip this candidate into the exact H1/H2 genericity failure the brief warns against.
4. Statistical-power risk: AI4SmallFarms covers only two countries and 62 tiles, limiting the power of a four-axis stratified ANOVA; conclusions may not generalise even within SEA.
5. Confound risk: any measured gain from the targeted fix could simply be domain-adaptation-from-seeing-local-data rather than the specific terrain/canopy conditioning mechanism — this is exactly what the naive-fine-tune ablation is designed to test, and if that ablation shows no meaningful difference, the entire novelty claim fails.

Kill criteria

1. Week 1-2: (1) Run frozen Delineate Anything v2 zero-shot on AI4SmallFarms test tiles and compute the four-axis stratified breakdown immediately; if performance does not vary meaningfully across DEM-slope and WorldCover tree-cover buckets (i.e. failure is uniform, or dominated by an axis such as resolution or cloud cover that adjacent papers have already fixed), the core premise is false and the project should pivot or stop.
2. Explicitly check whether AI4SmallFarms' actual footprint contains sufficient slope/canopy variance to stratify at all; if the region is essentially flat and canopy-sparse, either widen to a supplementary hand-labelled region in week 1 or abandon.
3. Set a standing arXiv watch on the FTW and Delineate Anything author groups; if a canopy/terrain-conditioned SEA-specific model is published during the project, pivot the delta immediately or reframe as an independent replication with a different, narrower angle.

Open / unverified items

- Whether AI4SmallFarms' actual Vietnam/Cambodia footprint contains sufficient slope and canopy-cover variance to support the four-axis stratified diagnosis the pitch assumes — this is the single most important open question and is also the week-1 kill-criterion check.
- Whether GTPBD-MM's geographic coverage or ablations already extend to SEA-like agroforestry/canopy-overlap conditions (only the abstract was reviewed in this pass, not the full method/results).
- Exact free-tier access terms and current licence string for AI4SmallFarms and Delineate Anything v2's GitHub release (not independently re-verified beyond search-result summaries).

- ESA WorldCover's current Planetary Computer access route was not independently re-fetched in this pass (based on well-established prior knowledge rather than a fresh check).
- A specific, sourced quantified-pain number for hectares or economic cost of unreliable field boundaries specifically in Vietnamese/Cambodian smallholder agroforestry or terraced systems.
- Whether a free tropical-forest PlanetScope/NICFI tier could usefully supplement AI4SmallFarms for canopy-overlap examples — considered but not verified, and treated as optional/skippable.

Key references

Delineate Anything v2: A Global Foundation Model for Field Delineation

arXiv preprint · 2026 · [arXiv:2607.19069](#)

The global model being diagnosed; source of the 100-country zero-shot benchmark this candidate decomposes into strata.

Fields of The World: A Machine Learning Benchmark Dataset for Global Agricultural Field Boundary Segmentation

AAAI · 2025 · [arXiv:2409.16252](#)

Establishes the multi-country zero-shot/fine-tuning evaluation convention this candidate's stratified protocol builds on.

GTPBD-MM: A Global Terraced Parcel and Boundary Dataset with Multi-Modality

arXiv preprint · 2026 · [arXiv:2604.12315](#)

Already validates the DEM+text terrain-conditioning recipe this candidate proposes to adapt — the closest direct methodological prior art, though not SEA-specific.

AI4SmallFarms: A Dataset for Crop Field Delineation in Southeast Asian Smallholder Farms

IEEE Geoscience and Remote Sensing Letters · 2023 · [IEEE Xplore 10278130](#)

The only verified, free, real-ground-truth SEA smallholder field-boundary dataset this candidate can actually build on — its coverage limits directly bound the project's feasibility.

Fields of the Planet: Field Boundary Mapping Beyond 10m

arXiv preprint · 2026 · [arXiv:2607.04449](#)

Already demonstrates that higher-resolution imagery, not terrain/canopy conditioning, substantially fixes the small-field/fragmentation failure mode — directly threatens this candidate's novelty if fragmentation turns out to dominate its own diagnosis.

Tackling fluffy clouds: robust field boundary delineation across global agricultural landscapes with Sentinel-1 and Sentinel-2 Time Series (PTAViT3D-CA)

arXiv preprint · 2024 · [arXiv:2409.13568](#)

Already validated on AI4SmallFarms itself, using SAR+optical fusion to fix cloud-cover-driven failure — a major real SEA confound this candidate's diagnosis must rule out before attributing failure to terrain/canopy.

PRUE: A Practical Recipe for Field Boundary Segmentation at Scale

arXiv preprint · 2026 · [arXiv:2603.27101](#)

Already runs a systematic 18-model robustness evaluation on the FTW benchmark, partially closing the general 'systematic diagnosis' novelty space this candidate occupies.

Generated from the source proposal JSON. Amber [uncertain] flags are carried over verbatim from the source and mark claims that were not independently verified.